

Crypto Report

Privacy-Preserving Sealed-Bid Auction - Cryptographic Implementation Report

Executive Summary

This report documents a privacy-preserving sealed-bid auction system that leverages advanced cryptographic techniques to ensure bid confidentiality while enabling fair winner determination. The system combines Pedersen commitments, zero-knowledge range proofs, additive secret sharing, and 3-party multi-party computation (MPC) to achieve its security goals.

System Architecture

Core Components

- **Pedersen Commitments Module:** Implements elliptic curve-based commitments for bid hiding
- **Zero-Knowledge Range Proofs Module:** Provides bitwise range proofs to validate bid bounds without revealing values
- **Multi-Party Computation Module:** Enables secure computation among three parties to find the maximum bid
- **Main Auction Module:** Orchestrates the auction flow integrating all cryptographic primitives

Cryptographic Primitives

1. Elliptic Curve Pedersen Commitments

Purpose: Allows bidders to commit to their bids in a binding yet hiding manner.

Construction:

A commitment is computed as: $C = bG + rH$

Where:

- b = bid value (secret)
- r = randomness/blinding factor (secret)
- G, H = independent generator points on the elliptic curve

Properties:

- **Hiding:** The commitment reveals no information about the bid value
- **Binding:** Once committed, the bidder cannot change their bid
- **Homomorphic:** Commitments can be added: $C_1 + C_2 = (b_1 + b_2)G + (r_1 + r_2)H$

2. Zero-Knowledge Range Proofs (Bitwise)

Purpose: Proves that a committed bid lies within the valid range $[b_{\min}, b_{\max}]$ without revealing the actual bid.

Protocol:

1. Compute adjusted value: $\text{adjusted} = \text{bid} - b_{\min}$ (must be ≥ 0)
2. Decompose adjusted into binary representation: $\text{adjusted} = \sum (2^i \times \text{bit}_i)$
3. Create Pedersen commitments for each bit: $B_i = \text{bit}_i \times G + r_i \times H$
4. For each bit commitment B_i , prove it opens to either 0 or 1 using a 2-way OR proof (Chaum-Pedersen style, made non-interactive via Fiat-Shamir transform)
5. Compute the sum: $S = \sum (2^i \times B_i)$
6. Compute: $C' = C - b_{\min} \times G$ and $D = C' - S = r_{\text{extra}} \times H$
7. Prove knowledge of r_{extra} using a Schnorr proof on D with respect to H
8. Verifier checks all bit proofs, the Schnorr proof, and validates: $C = b_{\min} \times G + S + D$

Characteristics:

- Proof size and computation are linear in bit-length (for max value 1000, approximately 10 bits required)

- Educational/demonstration implementation - production systems should use Bulletproofs or similar optimized range proofs
- Non-interactive and publicly verifiable

3. Multi-Party Computation (3-Party MPC)

Purpose: Enables three computational parties to jointly find the maximum bid without revealing individual bids to each other.

Secret Sharing: Uses additive secret sharing where a value v is split into three shares such that $v = \text{share}_1 + \text{share}_2 + \text{share}_3$

Supported Operations:

- **mpc_add:** Addition of two secret-shared values (locally computed)
- **mpc_sub:** Subtraction of two secret-shared values (locally computed)
- **mpc_mult:** Multiplication of secret-shared values (simplified protocol with communication)
- **mpc_compare:** Comparison of two secret-shared values using difference reconstruction
- **mpc_argmax:** Finds the index of the maximum among secret-shared values

Comparison Protocol:

1. Each server locally computes the difference between its shares of two values
2. Servers exchange these share differences
3. The difference (but not individual values) is reconstructed
4. The sign of the difference determines which value is larger

Auction Protocol Flow

Phase 1: Bid Commitment

1. Each bidder selects their bid b_i within the valid range $[b_{\min}, b_{\max}]$
2. Bidder generates random blinding factor r_i
3. Bidder computes commitment $C_i = b_i \times G + r_i \times H$

4. Bidder generates zero-knowledge range proof π_i for C_i
5. Bidder publishes (C_i, π_i)

Phase 2: Proof Verification

1. All parties verify each range proof π_i against commitment C_i
2. Invalid proofs result in bidder disqualification

Phase 3: Secret Sharing

1. Each bidder splits their bid b_i into three additive shares
2. Shares are distributed to three MPC computation servers
3. Each server receives one share from each bidder

Phase 4: Secure Maximum Finding

1. MPC servers execute the mpc_argmax protocol on secret-shared bids
2. Through iterative comparisons, servers determine the index of the winning bid
3. Servers reconstruct only the winner's ID and winning bid value
4. All other bids remain secret-shared and private

Phase 5: Result Announcement

1. The winner ID and winning bid are publicly announced
2. All losing bids remain confidential

Security Properties

Privacy Guarantees

- **Bid Confidentiality:** All losing bids remain hidden throughout the protocol
- **Commitment Hiding:** Pedersen commitments reveal no information about bid values
- **Zero-Knowledge:** Range proofs reveal only that bids are within valid bounds
- **MPC Security:** Secret sharing ensures no single party learns individual bids

Integrity Guarantees

- **Binding Commitments:** Bidders cannot change bids after commitment phase
- **Range Enforcement:** Zero-knowledge proofs ensure all bids are within valid bounds
- **Correct Computation:** MPC protocol ensures the true maximum is found

Threat Model

- **Honest-but-Curious MPC Servers:** Assumes servers follow protocol but may try to learn information
- **Malicious Bidders:** Range proofs prevent out-of-range bids
- **Collusion Resistance:** Requires at least one honest MPC server for privacy

Implementation Considerations

Performance Characteristics

- **Commitment Generation:** Fast elliptic curve operations (milliseconds)
- **Range Proof Generation:** Linear in bit-length, suitable for small-to-medium ranges
- **MPC Comparison:** Requires communication between servers, complexity grows with number of bidders
- **Verification:** Fast for all cryptographic proofs

Limitations and Future Improvements

- **Range Proof Efficiency:** Current bitwise approach is educational; production should use Bulletproofs for logarithmic proof size
- **MPC Simplification:** Multiplication protocol is simplified; production should use robust MPC frameworks
- **Server Trust:** Current design requires trusted MPC servers; could be enhanced with verifiable computation

- **Scalability:** Linear growth in comparisons; could optimize with tournament-style comparison trees

Recommended Production Enhancements

1. Replace bitwise range proofs with Bulletproofs for efficiency
2. Implement robust MPC protocol with malicious security (e.g., SPDZ)
3. Add verifiable computation to prove correct MPC execution
4. Implement distributed key generation for elliptic curve parameters
5. Add audit logging and dispute resolution mechanisms

Conclusion

This cryptographic implementation provides a solid foundation for privacy-preserving sealed-bid auctions. The combination of Pedersen commitments, zero-knowledge range proofs, and multi-party computation achieves the core security goals of bid privacy and auction integrity. While the current implementation serves educational purposes, the architecture can be enhanced with production-grade cryptographic libraries to create a fully deployable system suitable for real-world auction scenarios.